Soumodeep Karmakar

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Education

B.E. Electrical Engineering 2024 – 2028 (Expected)
Jadavpur University, Kolkata | CGPA: 7.5 (upto 4th sem) *Kolkata, India*

Skills

AI & Data Science: Machine Learning, Deep Learning, Generative AI, LLMs, Transformers, Mixture-of-Experts, Statistical Modeling, NLP, Semantic Analysis, Computer Vision, Multimodal Learning, RAG, Agentic AI
Frameworks & Cloud: PyTorch, scikit-learn, NumPy, Hugging Face, FastAPI, Pydantic AI, PostgreSQL, SQLite, React, Modal, AWS (EC2), Docker, Git
Languages & Core CS: Python, SQL, C/C++, JavaScript, TypeScript, OOP, Data Structures & Algorithms

Experience

Research Contributor Jun 2026 – Present
CMATER Lab, Jadavpur University (under Prof. Ram Sarkar) *Kolkata, India*

- Built a **cross-dataset, few-shot medical image segmentation** pipeline using **SAM, CLIP, and CoOp** to generalize across unseen medical imaging domains.
- Developed **CLIP-guided mask selection** and pseudo-labeling for domain shift, contributing to **two papers** (one accepted at *CVIP 2026*, one first-author, under review).

Projects

AI Customer Support | *FastAPI, PostgreSQL, Pydantic AI, React, TypeScript, Docker* |

- Built a **Customer Experience** Gmail automation platform with **FastAPI**, **PostgreSQL**, **Pydantic AI**, and Docker to triage inbound threads, extract structured issue metadata, and draft policy-grounded replies.
- Implemented **PostgreSQL full-text policy retrieval** and deterministic business-rule guardrails with connection pooling, stale-job recovery, and duplicate-reply protection.
- Shipped a **React + TypeScript** admin dashboard for human-in-the-loop ticket resolution and knowledge-base editing, validated with **39 automated tests** and an **18-case golden email eval suite**.

nanoMoE | 120M params (65.8M active) | *PyTorch, MoE, MLA, Modal* |

- Built a **sparse Mixture-of-Experts LLM from scratch** with **DeepSeek-style Multi-head Latent Attention (MLA)** for KV-cache compression, RoPE, and SwiGLU.
- Designed a **Top-2 noisy-gating router** with auxiliary load-balancing loss (four experts per layer); pre-trained on **FineWeb-Edu** on cloud H100 GPUs via Modal.

ReAct Agent Framework | *Python, FastAPI, Modal, SWE-bench* |

- Built a modular **ReAct agent framework** with a code-debugging agent (sandboxed shell execution, git patch export) and a chess agent that runs **programmatic lookahead search** inside an isolated container.
- Designed a **working-memory context compaction** engine; resolved SWE-bench Django task django-15368 while cutting peak prompt size by **86.2%** (35.7K → 4.9K tokens) and total tokens by **86.3%**.
- Added an offline test suite, transient retry handling, and cloud container sandboxing to isolate model-written code.

Durable Tool-Execution Runtime | *Python, asyncio, SQLite* |

- Built a crash-recoverable Python execution backend for AI agent tool calls with single-writer **SQLite WAL** persistence, bounded backpressure, argument-hashed idempotency, and **epoch-fenced worker leases**.
- Validated via **560 automated tests**, real process-kill (**SIGKILL**) crash-recovery chaos testing with zero lost jobs, and **2,115 jobs/s** sustained throughput (**3.63 ms** p99 admission latency).

Publications

- **Calibrated SAM-CLIP Pseudo-Labeling and Cross-Dataset CutMix for Few-Shot Medical Image Segmentation.** **S. Karmakar**, S. Pani, S. Ghosh, J. Banerjee, R. Pal, R. Sarkar. *Knowledge-Based Systems (Elsevier).* **Submitted (under review)**.
- **Beyond Spatial Prototypes: Joint Spatial–Spectral Prototype Learning for Few-Shot Medical Image Segmentation.** J. Banerjee, R. Pal, S. Ghosh, S. Pani, **S. Karmakar**, R. Sarkar. *CVIP 2026.* **Accepted**.

Achievements

AISEHack 2026 – Top 4 Team, IIIT Hyderabad (IBM Theme: Flood Detection) Apr 2026
Medha-Manthan 2025 – 1st Prize, IEEE Kolkata Chapter (assistive communication board) Sep 2025
Hackforge – 3rd Position, Srijan'25, CodeClub Jadavpur University Mar 2025